



Winding Tree

ERC827 Standard (ERC20 Extension)



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A few months ago, when I started working on smart contracts for the proof of concept of Winding Tree, I realized how limited the ERC20 token is, for our use case. For example, to book a hotel room with an ERC20 token, the sender (an OTA) needs to approve the outgoing transaction first, and then the receiver (a hotel) has to claim it. All this can be done in one tx if you use ether, but not with ERC20.

The problem is that ERC20 is like Bitcoin. The standard allows you just to transfer value, compare it with Ethereum where in addition you can execute a call in a transaction, which allows you to do a lot more!

After a while I came up with this simple solution that reuses the ERC20 code as much as possible, the new contract uses just three new functions.

```
contract ERC827 is ERC20 {  
    function approve(  
        address _spender, uint256 _value, bytes _data  
    ) public returns (bool) {  
        require(_spender != address(this));  
        super.approve(_spender, _value);  
    }  
}
```

```

        require(_spender.call(_data));
        return true;
    }

    function transfer(
        address _to, uint256 _value, bytes _data
    ) public returns (bool) {
        require(_to != address(this));
        super.transfer(_to, _value);
        require(_to.call(_data));
        return true;
    }

    function transferFrom(
        address _from, address _to, uint256 _value, bytes _data
    ) public returns (bool) {
        require(_to != address(this));
        super.transferFrom(_from, _to, _value);
        require(_to.call(_data));
        return true;
    }
}

```

This is it, these **30 lines of code** allow us to transfer value and data, so now we can:

- Build more complex apps over tokens
- Increase the token utility
- Execute more complex functions in token transfers/approvals
- Store data on chain in token transfers/approvals

The new standard is the result of months of work, research and collaboration with Zeppelin and their active community.

Special thanks to **Santiago Palladino**, **Facu Spagnuolo** and **Francisco Giordano** from **Zeppelin Solutions** that helped me a lot to get the standard finished and ready to be presented to the community.

Now the ERC827 token is available to be used in zeppelin-solidity 1.6.0 ! 

EIP — ERC827

ERC827 LifToken Implementation

ERC827 Interface in zeppelin-solidity

ERC827 Standard Token Implementation in zeppelin-solidity

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	verification				contract configuration	
Registration / Invalidation	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted		-Approve minting of new LIFs to the "customers rewards" fund	
Verification		-Verifies suppliers or retailers -Invalidates suppliers or retailers -Pays verification fee with LIF to the DAO from its deposit balance			-Validates and verifies verifiers -Invalidates verifiers	
Deals creation			-Creates a deal on behalf of a customer and pays an offer price	-Creates a deal and pays an offer price		
Deals rejection	-Pays tx expenses -Pays a proper amount of LIF from its deposit as a rejection penalty			-Receives a full refund of the deal funds	-Receives "rejection fee" from the supplier	
Deals cancellation	-Receives fullpart of	-Receives part of the	-Receives part of the	-Pays tx expenses	-Receives part of the	

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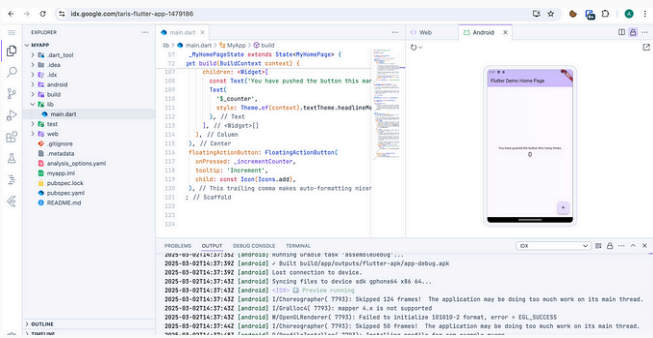
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